Unit 2.1 Printables: DNA to Proteins

Everything students glue in, annotate, or use on paper, in day order. Cut each sheet on the dashed line and glue it into the matching “Glue in” box on that day's notebook setup page (Unit21_Student_Notebook.pdf). Print on white paper, black ink only. Answers: Unit21_Printables_Teacher_Key.pdf.

Day	Printable	Use	Copies
Day 1	Siamese cat data	Glue in · annotate	1 per student
Day 1	Reading: DNA, genes, and chromosomes	Glue in · annotate	1 per student
Day 2	Reading: What is DNA made of?	Glue in · annotate	1 per student
Day 3	Base key and code table	Lab table card	1 per pair
Day 4	Card sort: what can one letter do?	Glue in · write on it	1 per student
Day 5	Same DNA, different jobs	Glue in · write on it	1 per student
Day 6	Case data	Glue in · annotate	1 per student
Day 7	Write It: two Siamese kittens	Glue in · annotate	1 per student
Day 3	Gummy DNA lab sheet	Lab sheet (full page)	1 per pair

Days 2 and 8 need nothing else on paper. The notebook setups for each lesson are in Unit21_Student_Notebook.pdf.

UNIT 2.1 · DAY 1 · GLUE IN · ANNOTATE

Siamese cat data

Circle the 4 dark body parts. Write W next to the warm parts and C next to the cool parts.

Data 1: Temperature and fur color

Body part of an adult Siamese cat	Skin temperature (°C)	Fur color
Back (trunk)	36	Cream
Belly	36	Cream
Upper leg	34	Cream
Ear	28	Dark brown
Nose and face mask	30	Dark brown
Paw	28	Dark brown
Tail tip	27	Dark brown

Simplified class data set. Real temperatures vary.

Data 2: Litters of kittens

Litter	Mother	Father	Dark-pointed (Siamese pattern) kittens	Solid-colored kittens
1	Siamese pattern	Siamese pattern	5	0
2	Siamese pattern	Siamese pattern	4	0
3	Solid	Solid	1	4
4	Solid	Solid	2	3

Simplified class data set modeled on real litter patterns.

UNIT 2.1 · DAY 1 · GLUE IN · ANNOTATE

Reading: DNA, genes, and chromosomes

Annotate: underline each key word the first time you see it (DNA, gene, protein, chromosome). Draw an arrow from DNA to gene to protein. Put a ? next to one thing you still wonder.

Every cell in a cat stores instructions in **DNA**. A **gene** is a section of DNA with the instructions for one **protein**. Proteins do the jobs of the cell, like making the dark pigment in fur.

DNA is packed into **chromosomes**. A chromosome is one long DNA molecule packed with proteins, and it holds many genes. A kitten gets one set of chromosomes, and so one set of genes, from each parent. That is why two solid-colored parents can still have a Siamese kitten: each parent can pass on a changed copy of the gene.

Big question: How can a difference in DNA change which parts of a cat are dark?

UNIT 2.1 · DAY 2 · GLUE IN · ANNOTATE

Reading: What is DNA made of?

Annotate: draw the shape next to each part (pentagon = sugar, circle = phosphate, rectangle = base). Box the base-pairing rule.

DNA is a **double helix**, shaped like a twisted ladder. It is built from **nucleotides**. Each nucleotide has three parts: a **sugar**, a **phosphate**, and a **base**.

The sugars and phosphates make the sides of the ladder (the backbone). The bases make the rungs. There are four bases: A, T, C, and G. **A pairs with T. C pairs with G.**

The order of the bases is the message. A cat's DNA has billions of bases, and a gene is one stretch of them.

Solid cat: A T G G G A C T G T T T

Siamese cat: A T G A G A C T G T T T *(made-up letters that show the idea)*

UNIT 2.1 · DAY 3 · LAB TABLE CARD

Base key and code table

Keep this card on your lab tray. Read the copy three letters at a time.

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Pairing rule: A with T, C with G.

Gummy DNA Lab: Build, Copy, and Read a Gene

Name: _____ **Partner:** _____

Question: How does a cell's copy of a gene become a chain of amino acids, and what happens if one letter of DNA changes?

You will build a short model of the start of a cat's gene for a pigment-making protein. (The letters are an illustration, not the real gene.)

Materials (per pair)

- 2 licorice strands (the backbone; the sugars and phosphates)
- Candy “bases”: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G) gummies, plus 2 extra of each color
- Toothpicks (to attach bases to the backbone and to hold pairs together)
- Paper plate or tray, this sheet, the code table

Safety: Keep materials clean. Do not eat the lab materials until your teacher says you may. Toothpicks are sharp. Tell your teacher about any food allergy. If candy is not used, use colored beads or paper circles.

Base key and code table

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Part A: Build the template strand (6 min)

This is the strand of the gene that the cell uses as a template. Attach the bases to one licorice strand in order from left to right:

T A C C C T G A C A A A

Write the color order you built: _____

Part B: Build the partner strand and make the ladder (8 min)

Build the partner strand on the second licorice strand by pairing each base (A with T, C with G). Write the partner sequence you built:

_ _ _ _ _ _ _ _ _ _ _ _

1. Join the two strands with toothpicks between pairs. Gently twist the ladder. Have your teacher check it before you continue. ☐ Checked
2. Which strand carries the “coding” letters (the same as the cell's message copy)? How do you know?

Part C: Copy the message, then read it (10 min)

The cell does not use the DNA directly to make a protein. It **copies the gene's message** from the template strand, and then the **copy is read** to link amino acids. The DNA stays safe.

1. Pull the two strands apart. Keep the template strand. Build a *new* strand by pairing with the template. This is the **copy**. Write its sequence: ___ ___ ___ ___
2. Read the copy three letters at a time with the code table. Write the amino acid chain: _____ – _____ – _____ – _____
3. Line up one bead or paper tag per amino acid on the plate in order. This chain would fold to make a protein.
4. Explain the path in your own words, starting with the DNA:
   _____

Part D: Change one letter (7 min)

1. On your template strand, replace the 4th base (the first C of the second triplet) with a T (swap a green gummy for a yellow one). The template now reads **TAC TCT GAC AAA**.
2. Build the new copy by pairing with the changed template. Sequence: ___ ___ ___ ___
3. Read it with the code table: _____ – _____ – _____ – _____
4. Compare with Part C. How many amino acids changed? Which one?
5. Predict: Could a protein built from this changed chain work differently? Explain.

Connect to the Siamese cat

1. In a Siamese cat, a one-letter change in the gene for a pigment-making protein swaps one amino acid. In your own words, what did your one-letter change do to the chain in the model?
2. Name two ways your gummy model is different from real DNA.

Clean up

Put toothpicks and candy in the trash or the labeled container. Wipe your table. Wash your hands.

UNIT 2.1 · DAY 4 · GLUE IN · WRITE ON IT

Card sort: what can one letter do?

For each card, write 1, 2, or 3 and a reason. Card B is done for you. 1 = changes an amino acid · 2 = changes whether the protein is made · 3 = no noticeable change

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Reason bank: No gene is involved. · The protein is not made. · The amino acid stays the same. · The amino acid changes from ___ to ___.

Card	What changed	1, 2, or 3	Reason
A	Siamese cat: GGA becomes AGA in the pigment gene.		
B	GAG becomes GAA.	3	The amino acid stays the same.
C	CTG becomes CCG.		
D	A change in DNA with no gene and no switch.		
E	A change in a gene's switch, so the gene is never turned on.		
F	Human gene: GAG becomes GTG.		

UNIT 2.1 · DAY 5 · GLUE IN · WRITE ON IT

Same DNA, different jobs

Highlight what is the same in cells P and Q. Then fill in the temperature table. We do the ear together.

Two kinds of cells

Cell type	Genes in the cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Where does the changed protein work? (cooler than 33 °C)

Body part	Skin temperature (°C)	Cooler than 33 °C? Protein works? (yes or no)	Fur color (dark or cream)
Ear	28	*yes (we do this one together)*	*dark*
Nose	30		
Paw	28		
Tail tip	27		
Upper leg	34		
Back	36		
Belly	36		

UNIT 2.1 · DAY 6 · GLUE IN · ANNOTATE

Case data

In each case, circle the DNA change, underline the protein change, and box the effect on the body.

Case 1: Siamese and solid cats

	Solid cat	Siamese cat
Gene for the pigment-making protein	Usual sequence	One letter different
Amino acid chain	Usual chain	One amino acid different
Where the protein works	At every body temperature	Only in skin cooler than about 33 °C
Fur color	Dark all over	Dark ears, nose, paws, tail; cream body
Temperature of ear / back	28 / 36 °C	28 / 36 °C

Case 2: A human gene

Person	Gene	Protein	Symptoms
1	Normal	Works	None
2	One base changed	One amino acid different; works poorly	Yes

UNIT 2.1 · DAY 7 · GLUE IN · ANNOTATE

Write It: two Siamese kittens

Circle the two numbers you will use as evidence. Then check off each part of your model as you add it.

Two Siamese kittens are raised from birth, one in a warm room and one in a cool room. At 12 weeks:

	Kitten raised warm	Kitten raised cool
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Model checklist

☐ DNA with the gene ☐ the one-letter change ☐ the amino acid chain with one amino acid different ☐ the protein working only in cool skin ☐ the dark points